Problem 1 (AMO 1983/2)

**Sketch:** For contradiction, let the roots be $r_1, \ldots r_5$. Using Vieta's,

$$2a^2 < 5b$$

$$\Leftrightarrow 2\left(-\sum_{cyc} r_1\right)^2 < \frac{5\sum_{sym} r_1 r_2}{12}$$

$$\Leftrightarrow \left(\sum_{cyc} r_1\right)^2 < \frac{5}{24}\sum_{sym} r_1 r_2 \ldots$$

Prove impossible by Muirhead.

Problem 8 (IMO 2000/2)

We let $a = \frac{x}{y}; b = \frac{y}{z}; c = \frac{z}{x}$

$$\left(a - 1 + \frac{1}{b}\right)\left(b - 1 + \frac{1}{c}\right)\left(c - 1 + \frac{1}{a}\right) \leq 1$$

$$\Leftrightarrow \left(\frac{x}{y} - 1 + \frac{z}{y}\right)\left(\frac{y}{z} - 1 + \frac{x}{z}\right)\left(\frac{z}{x} - 1 + \frac{y}{x}\right) \leq 1$$

$$\Leftrightarrow (x - y + z)(y - z + x)(z - x + y) \leq xyz$$

$$\Leftrightarrow \sum_{sym} x^2 y - \frac{\sum_{sym} x^3}{2} - 2xyz \leq xyz$$

$$\Leftrightarrow \sum_{sym} x^2 y \leq (x^3 + xyz) + (y^3 + xyz) + (z^3 + xyz)$$

By AM-GM,

$$\Leftarrow \sum_{sym} x^2 y \leq \sum_{sym} x^2 y^{\frac{1}{2}} z^{\frac{1}{2}}$$

This is true by Muirhead.

Problem 9 (Taiwan Quiz 2014)

$$3(a+b+c) \geq 8\sqrt[3]{abc} + \sqrt[3]{\frac{a^3+b^3+c^3}{3}}$$

$$\Leftrightarrow \frac{a+b+c}{3} \geq \frac{8}{9}\sqrt[3]{abc} + \frac{\sqrt[3]{\frac{a^3+b^3+c^3}{3}}}{9}$$

By Power-Mean on the RHS,

$$\Leftarrow \frac{a+b+c}{3} \geq \sqrt[3]{\frac{\left(\frac{a^3+b^3+c^3}{3}\right)}{9} + \frac{8}{9}(abc)}$$

$$\Leftrightarrow (a+b+c)^3 \geq (a^3+b^3+c^3) + 24abc$$

This is true by Muirhead.

Problem 13 (Iranian Olympiad)

By Cauchy,

$$\left(\sum_{cyc}\sqrt{x-1}\right)^2 \leq \left(\sum_{cyc} 1-\frac{1}{x}\right)\left(\sum_{cyc} x\right)$$

$$\Leftrightarrow \sum_{cyc}\sqrt{x-1} \leq \sqrt{x+y+z}$$

This is what we desire.

Problem 15 (IMO 2012/2)

$$(1+\mathrm{a}_2)^2(1+\mathrm{a}_3)^3 \ldots (1+\mathrm{a}_\mathrm{n})^n > n^n$$

$$\Leftrightarrow (1+\mathrm{a}_2)^2\left(\frac{1}{2}+\frac{1}{2}+\mathrm{a}_3\right)^3 \ldots \left((n-1)\left(\frac{1}{n-1}\right)+\mathrm{a}_\mathrm{n}\right)^n > n^n \qquad (a)$$

Note that, by AM-GM,

$$\frac{\frac{1}{k-1}(k-1)+a_k}{k} \geq \sqrt[k]{\left(\frac{1}{k-1}\right)^{k-1} a_k}$$

$$\Leftrightarrow \left(\frac{1}{k-1}(k-1)+a_k\right)^k \geq k^k\left(\frac{1}{k-1}\right)^{k-1} a_k$$

We apply this to every term on the LHS. Note that since we cannot have $a_k = \frac{1}{k-1}$ for all $k$, we can never reach the equality case.

$$(a) \Leftarrow \left(\frac{2^2}{1^1}a_2\right)\left(\frac{3^3}{2^2}a_3\right)\dots\left(\frac{n^n}{(n-1)^{n-1}}a_n\right) \geq n^n$$

$$\Leftrightarrow n^n \geq n^n$$

<u>Problem 16 (ELMO 2003, Po-Ru Loh)</u>

We claim that,

$$\frac{1}{x+1} \leq \frac{1}{4}\left(\frac{1}{x^2-1}\right) + \frac{1}{4}$$

Proof:

$$\Leftrightarrow x - 1 \leq \frac{1}{4} + \frac{x^2-1}{4}$$

$$\Leftrightarrow 0 \leq x^2 - 4x + 4$$

$$\Leftrightarrow 0 \leq (x-2)^2$$

Using this,

$$\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} \leq 1$$

$$\Leftarrow \frac{1}{4}\left(\frac{1}{x^2-1}\right) + \frac{1}{4}\left(\frac{1}{y^2-1}\right) + \frac{1}{4}\left(\frac{1}{z^2-1}\right) + \frac{3}{4} \leq 1$$

$$\Leftarrow \frac{1}{x^2-1} + \frac{1}{y^2-1} + \frac{1}{z^2-1} = 1$$

This is given.

<u>Problem 18 (IMO 2004/4)</u>

**Non-formally written roadmap**

WLOG, let $t_1 \leq t_2 \leq \cdots \leq t_n$.

$$t_i, t_j, t_k \text{ are the sides of a triangle for } 1 \leq i < j < k \leq n$$

$$\Leftrightarrow t_i + t_j > t_k \text{ for all } 1 \leq i < j < k \leq n$$

Since $t_1 \leq t_2 \leq \cdots \leq t_n$,

$$\Leftarrow t_1 + t_2 > t_n$$

For the sake of contradiction, let $t_1 + t_2 < t_n$.

Note that,

$$n^2 + 1 > (t_1 + t_2 + \cdots + t_n)\left(\frac{1}{t_1} + \frac{1}{t_2} + \cdots + \frac{1}{t_n}\right)$$

$$\Leftrightarrow n^2 + 1 > n + \sum_{1 \le i < j \le n} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i}\right)$$

$$\Leftrightarrow n^2 + 1 > n + \sum_{1 \le i < j \le n; (i,j) \ne (1,n); (i,j) \ne (2,n)} \left(\frac{t_i}{t_j} + \frac{t_j}{t_i}\right) + \left(\frac{t_1}{t_n} + \frac{t_n}{t_1}\right) + \left(\frac{t_2}{t_n} + \frac{t_n}{t_2}\right)$$

By AM-GM,

$$\Leftarrow n^2 + 1 > n + 2\left(\frac{n^2 - n}{2} - 2\right) + \left(\frac{t_1}{t_n} + \frac{t_n}{t_1}\right) + \left(\frac{t_2}{t_n} + \frac{t_n}{t_2}\right)$$

$$\Leftrightarrow 5 > \left(\frac{t_1}{t_n} + \frac{t_n}{t_1}\right) + \left(\frac{t_2}{t_n} + \frac{t_n}{t_2}\right)$$

This is impossible using the fact that $t_1 + t_2 < t_n$. So we are done.

Problem 22 (MOP 2011 R4.2)

$$\sum_{cyc} \sqrt{\frac{a^3 + b^3}{a + b}} + 9\sqrt[3]{abc} \le 12$$

$$\Leftrightarrow \sum_{cyc} \sqrt{a^2 - ab + b^2} + 9\sqrt[3]{abc} \le 4(a + b + c) \qquad (a)$$

Note that by Muirhead, since $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \prec \left(\frac{1}{2}, \frac{1}{2}, 0\right)$,

$$9\sqrt[3]{abc} \le 3\left(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}\right)$$

$$(a) \Leftarrow \sum_{cyc} \sqrt{a^2 - ab + b^2} \le 4(a + b + c) - 3\left(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}\right)$$

$$\Leftrightarrow \sum_{cyc} \sqrt{a^2 - ab + b^2} \le \sum_{cyc} 2a - 3\sqrt{ab} + 2b$$

$$\Leftarrow \sqrt{a^2 - ab + b^2} + 3\sqrt{ab} \le 2(a + b)$$

$$\Leftrightarrow \left(\frac{\sqrt{a^2 - ab + b^2} + 3\sqrt{ab}}{4}\right)^2 \le \left(\frac{a + b}{2}\right)^2 \quad (a)$$

By Power-Mean,

$$\left(\frac{\sqrt{a^2 - ab + b^2} + 3\sqrt{ab}}{4}\right)^2 \leq \frac{a^2 - ab + b^2 + 3ab}{4}$$

This is (a) rewritten, so we are done.